January 2026

Czech Bank Analysis

Financial Analysis using Berka_dataset | SQL

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1. Project Context

The study need to meet the following objectives

1 Account and demographic analysis

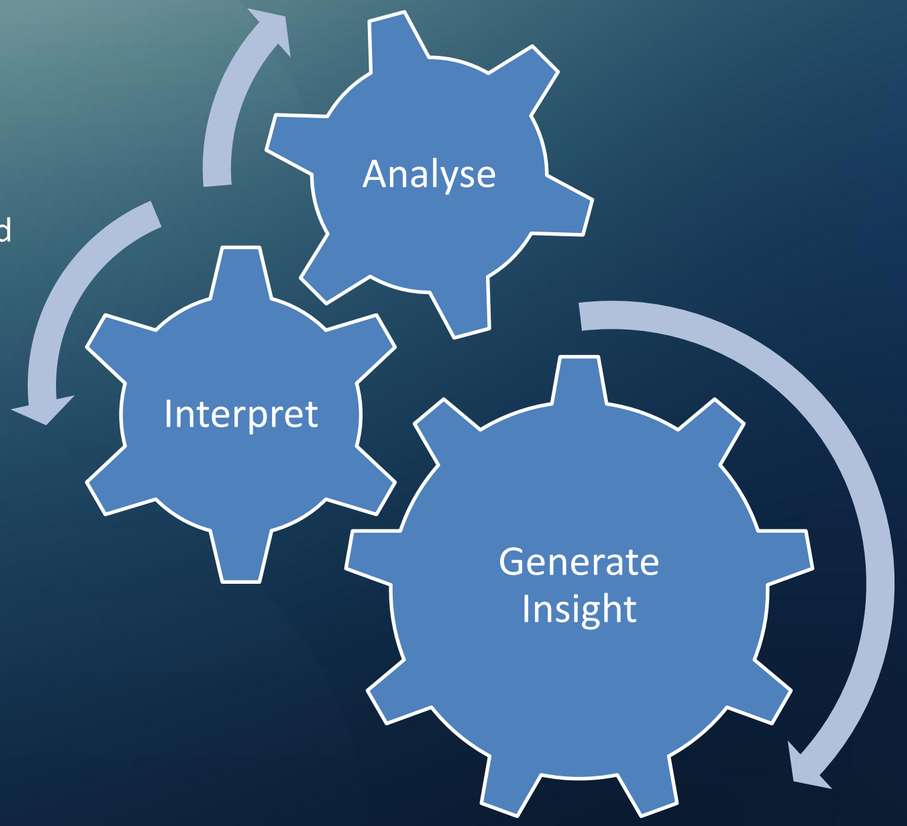
Understand who are Czech Bank's customers; their demographics, location and account types

2 Conduct transactional analysis

Analyse 1million + transactions to understand spending behaviour and account activity and trends over time

3 Conduct Loan and Credit card analysis

Analyse the bank's loan book. Understand the distribution of loan sizes, durations, repayment statuses, and the profile of borrowers.



The following dataset was analysed to meet the outlined objectives

Financial analysis was conducted on a dataset from the Czech bank, named the Berka_dataset. The dataset contained anonymised transactional data from Czech bank from 1993 – 1999, including 1 million+ transactions, 5369 clients and 682 loans over 77 districts.

The dataset has 8 tables:

- **Account:** Account information
- **Clients:** Client birth dates and districts
- **Disposition:** Distinguishes between owner and disponent account users
- **Permanent orders:** Transactions showing money deducted from accounts
- **Transactions:** Includes both credit and debit transactions
- **Loans:** Loans that were granted
- **Credit card:** Credit cards that were issued
- **Demograph:** Demographic information about each districts

Source: *https://webpages.charlotte.edu/mirsad/itcs6265/group1/domain.html*

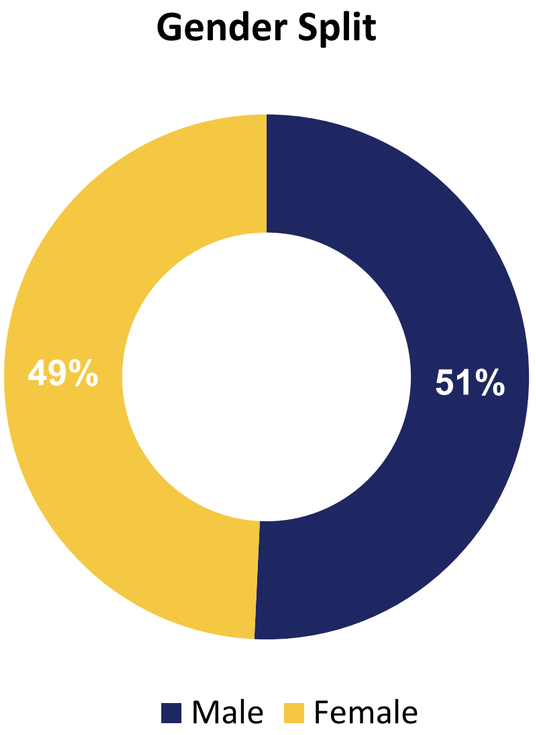
The tables were analysed on MySQL, view the workings here: *https://github.com/maselash/Financial-Analysis-with-Berka_dataset*

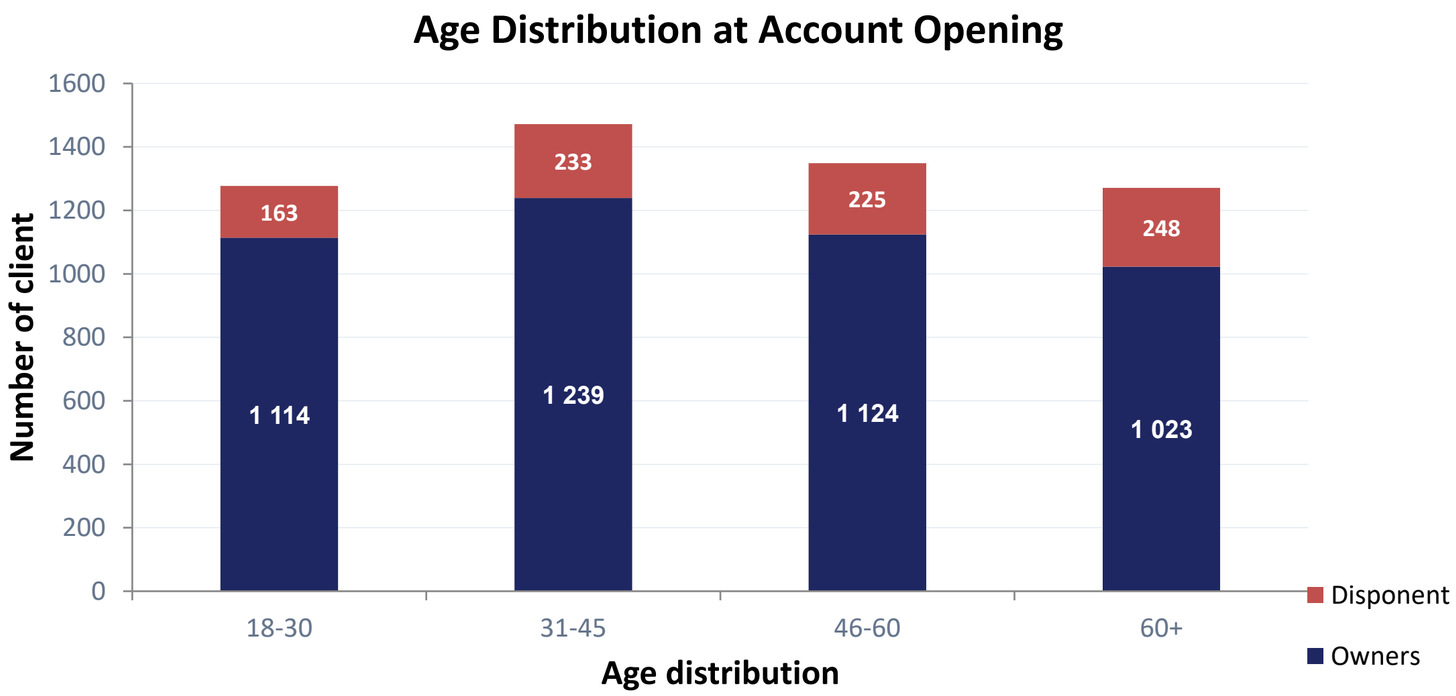
Disclaimer: This report is written based on 1999 data, although the data may be old the purpose is to showcase my financial analytical skills

2. Account and transactional analysis

The Czech bank's customers are almost evenly split by gender, with the majority falling between the ages of 31 and 60 years old

- The total number of clients comprises both account owners and disponent users.
- A disponent is an authorized user who has been granted permission by the account owner to access and transact on a bank account.
- While they are able to manage funds, they are unable to apply for loans or withdraw money from the account.

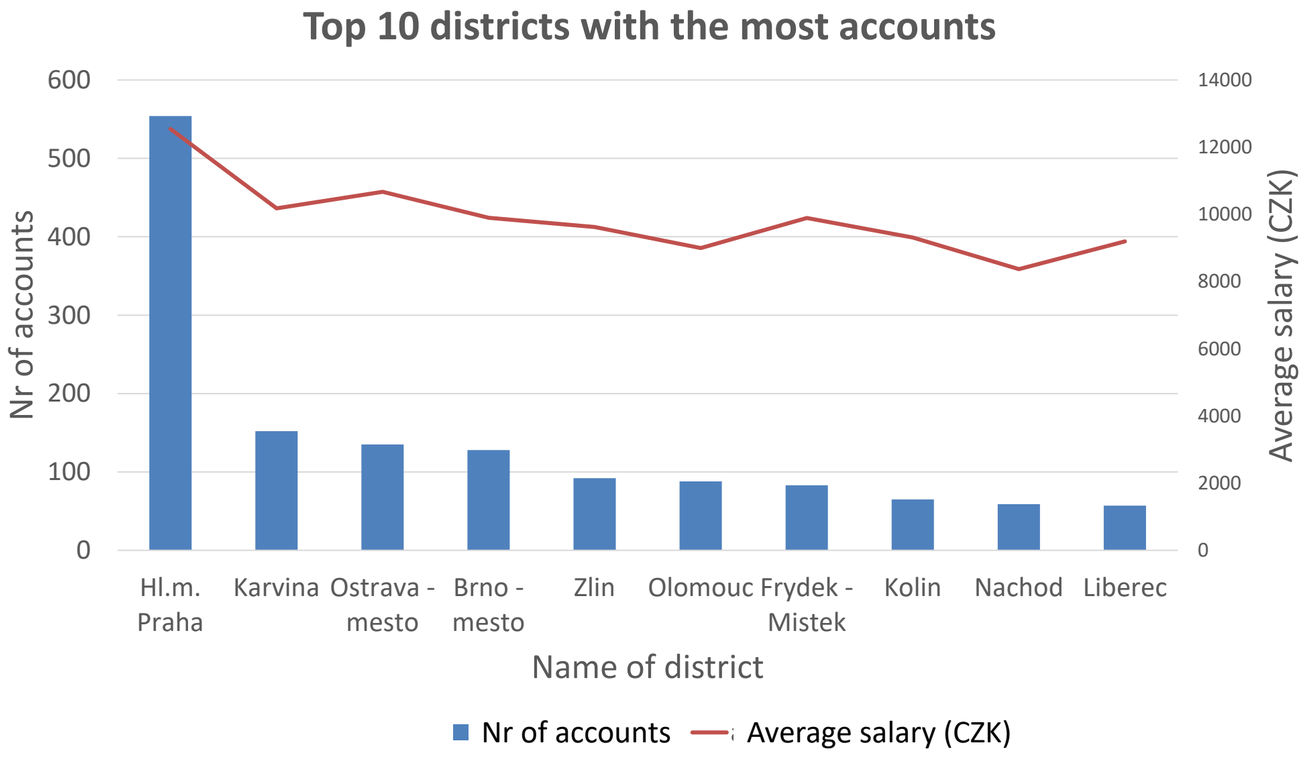


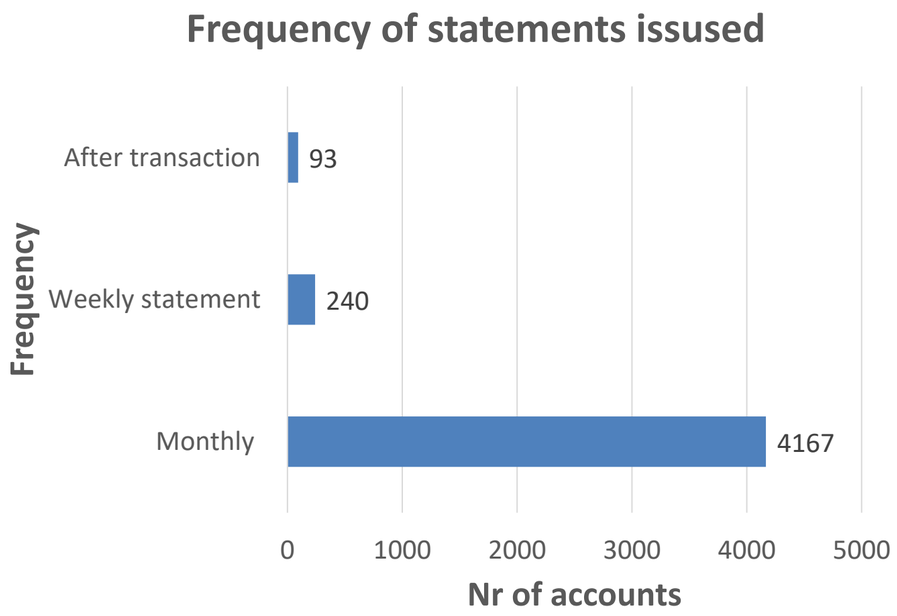


Reference tables:
Account – 4500 rows
Clients – 5369 rows
Disposition – 5369 rows

Hlavní město Praha is the district with the highest number of accounts and the highest average salary compared to other districts

- Hlavní město Praha is the capital city of Prague and the largest city in the Czech Republic, which explains why residents earn a higher average salary and hold the most accounts compared to other districts.
- Majority of accounts have their statements issued monthly. Explored in the subsequent slides, this influences their transactional behaviour and how they manage their money.





Reference tables:
Account – 4500 rows
Demograph – 77 rows

Majority of Czech Bank’s clients are middle aged and working towards meeting specific financial goals

Opportunity to tailor products based on their life-stage to help them meet their financial goals

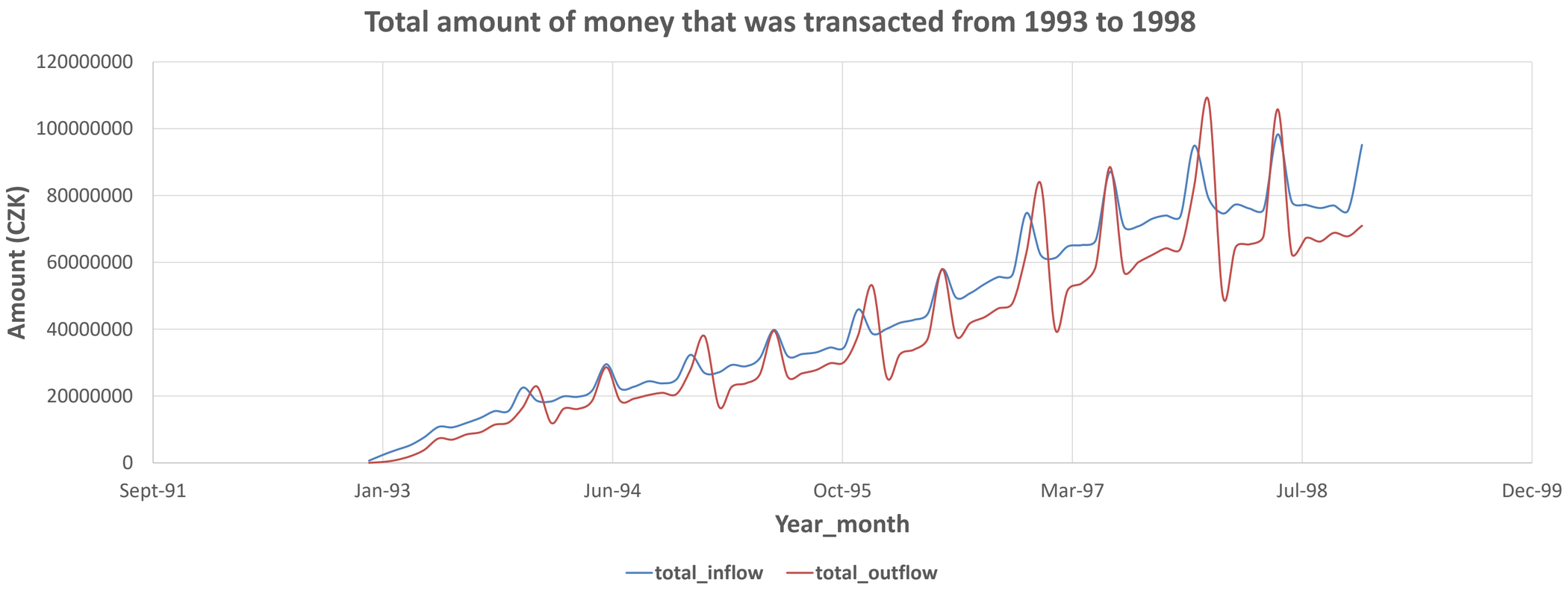
- The bank's core clients are mid-career earners with stable financial needs such as mortgages, retirement planning, insurance, and credit.
- Products should help them reach their financial goals, help them build and manage their wealth
- Medium to long-term loans will be crucial in helping them meet these goals.

Untapped banking potential in multiple regions

- Prague is the bank's primary revenue engine.
- However, the other districts represents an underbanked opportunity.
- The bank should investigate if they are under-serving these districts
- There is opportunity to open more branches and reach out to potential clients via various marketing channels to communicate the value they offer

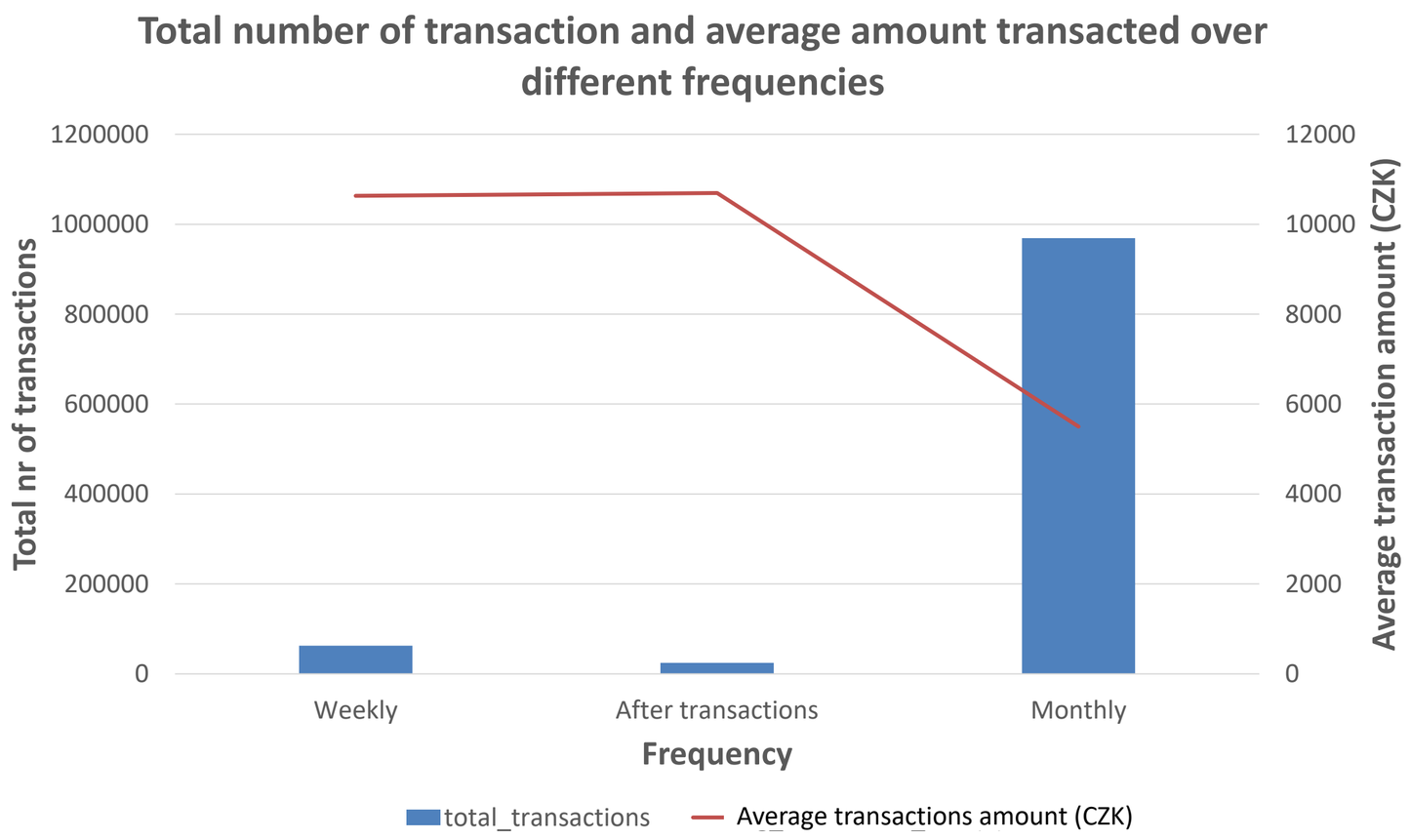
There is a consistent and steady increase in the total money transacted from 1993 to 1998

- Money **transacted out peaks between December and January**, as a result of the festive season.
- Money **transacted in is at its highest from February to June and from July to November**.
- Money coming in and going out are equal in June.



Reference tables:
Transactions – 1 million + rows

Accounts that issue monthly statements have the highest number of transactions but the average amount of money transacted is the lowest



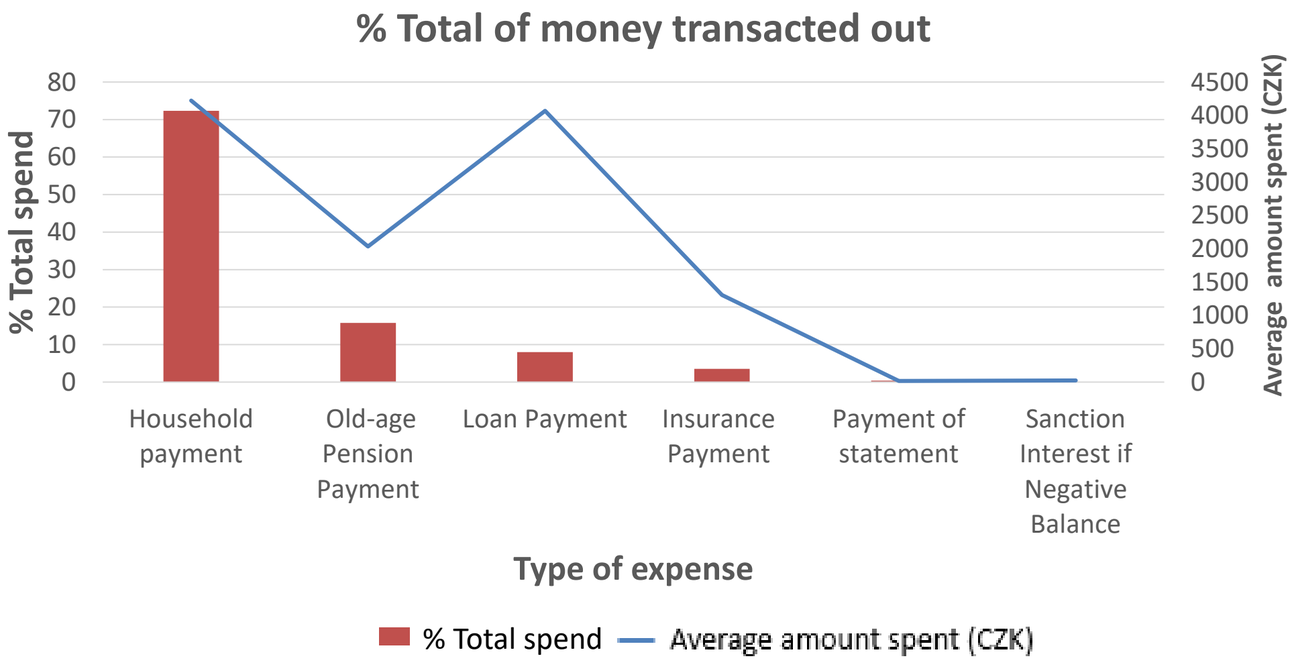
- It was interesting to note that accounts with statements issued weekly and after each transaction had the lowest total number of transactions.
- However average amount of money transacted was the highest.
- These account holders can be considered high-value clients, who prefer to be notified of their transactions regularly.

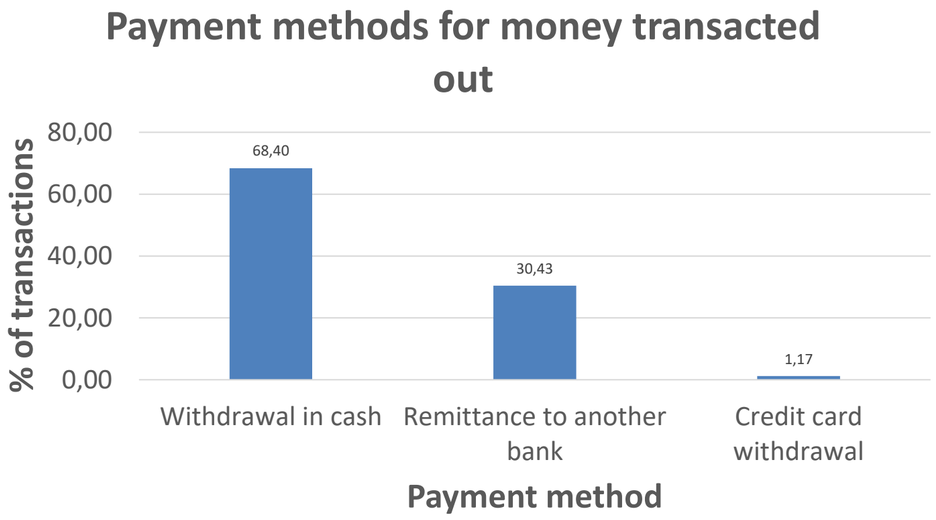
Let us unpack where their money is going…

Reference tables:
Account – 4500 rows
Transactions – 1 million + rows

Most transactions are towards household items; however, the average amount of money spent on household payments and loan repayments is the same

- Most transactions were towards household items (rent, electricity, water, etc.), but the average amount of money spent was the same as that of loan repayments.
- Clients settled their expenses primarily through cash withdrawals and bank transfers.





Reference tables:
Transactions – 1 million + rows

The transactional volume is steadily growing, and customers are cash heavy but there is low digitisation

This is a characteristic of Central/Eastern European banking in the 1990s

High Value clients have weekly and after transaction statements issued

- Weekly and After-transaction account holders transact far less frequently but move nearly 2x the average amount per transaction.
- These are likely business owners, high net-worth individuals.
- The bank should consider giving them a differentiated, premium service tier

Opportunity to increase loan and credit facilities for clients

- 72% of all outflow transactions are household payments (rent, utilities, etc.), followed by pension payments (16%) and loan repayments (8%).
- Credit card usage accounts for only 1.2% of outflow transactions
- There is opportunity to loan money and offer credit cards to clients between the ages 31-60.
- They are most likely to qualify and afford monthly repayments

However…

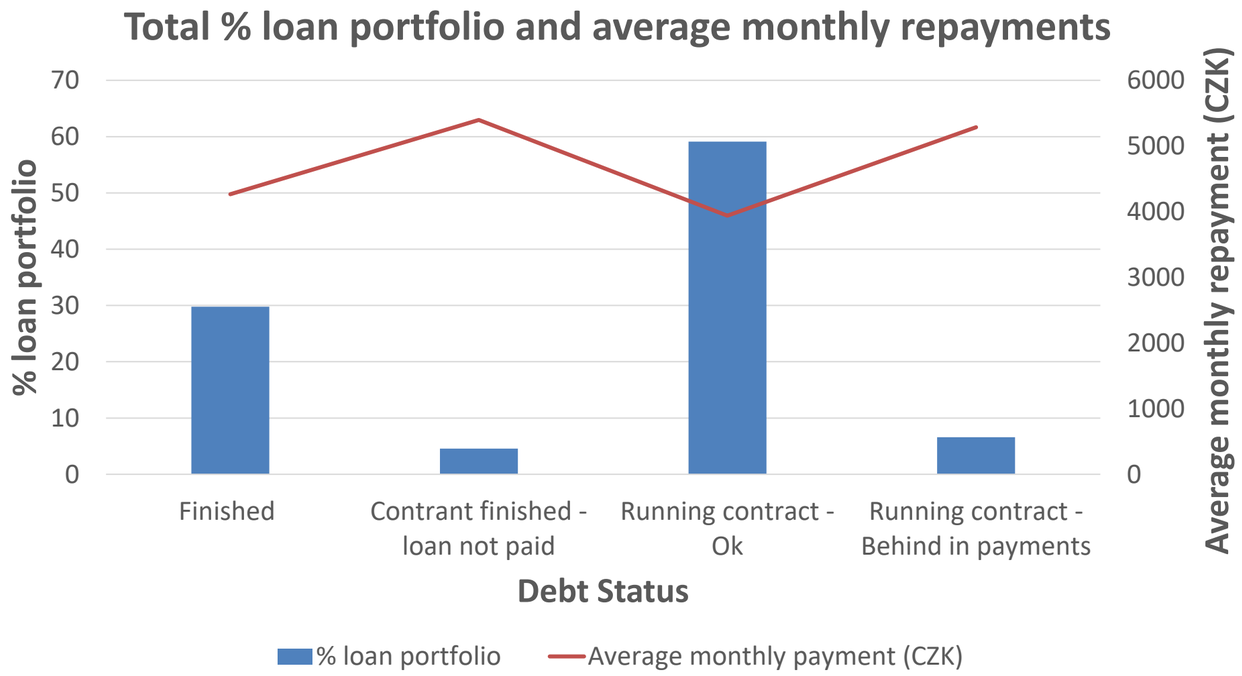
There is increased credit risk during festive seasons

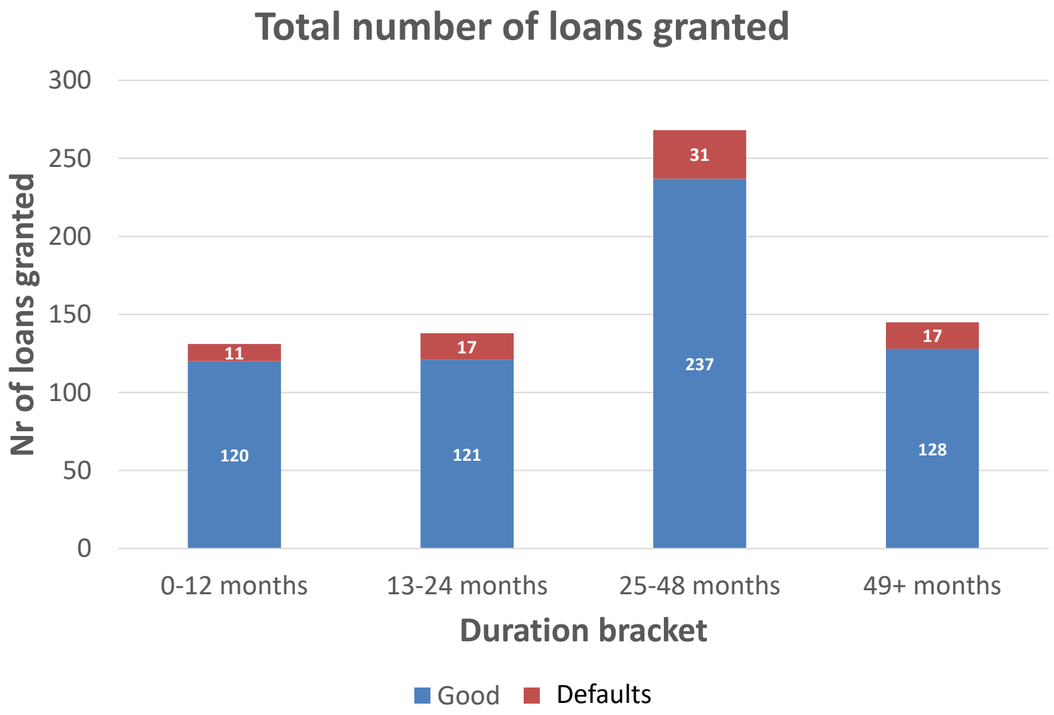
- A consistent pattern of more money be transacted during holiday season → this creates credit risk
- This should inform how credit cards/loans are granted during these season

3. Loan and credit card analysis

Defaulted loans have higher monthly repayments compared to those that are repaid on time

- Most loans granted by the bank are in good standing "*Running contract – ok",* however as the bank grows its loan portfolio, it is crucial to understand why clients default on their loans.
- Loans with a duration of over 24 months have the highest default rate therefore, a balance between loan duration and repayment amounts must be stuck to reduce the risk of default.

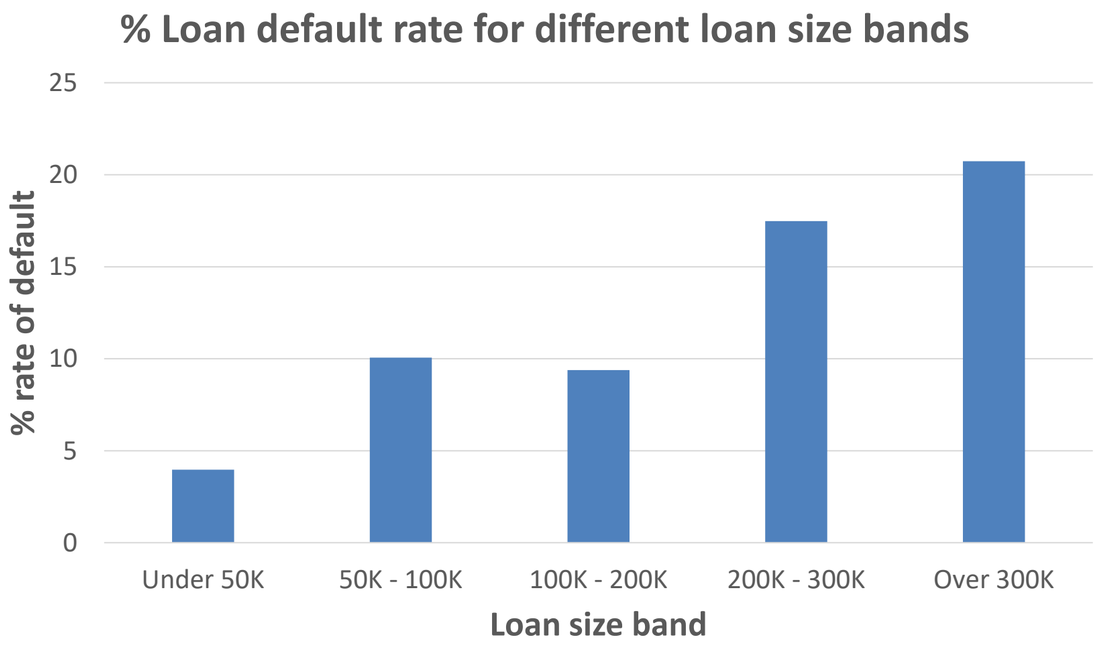


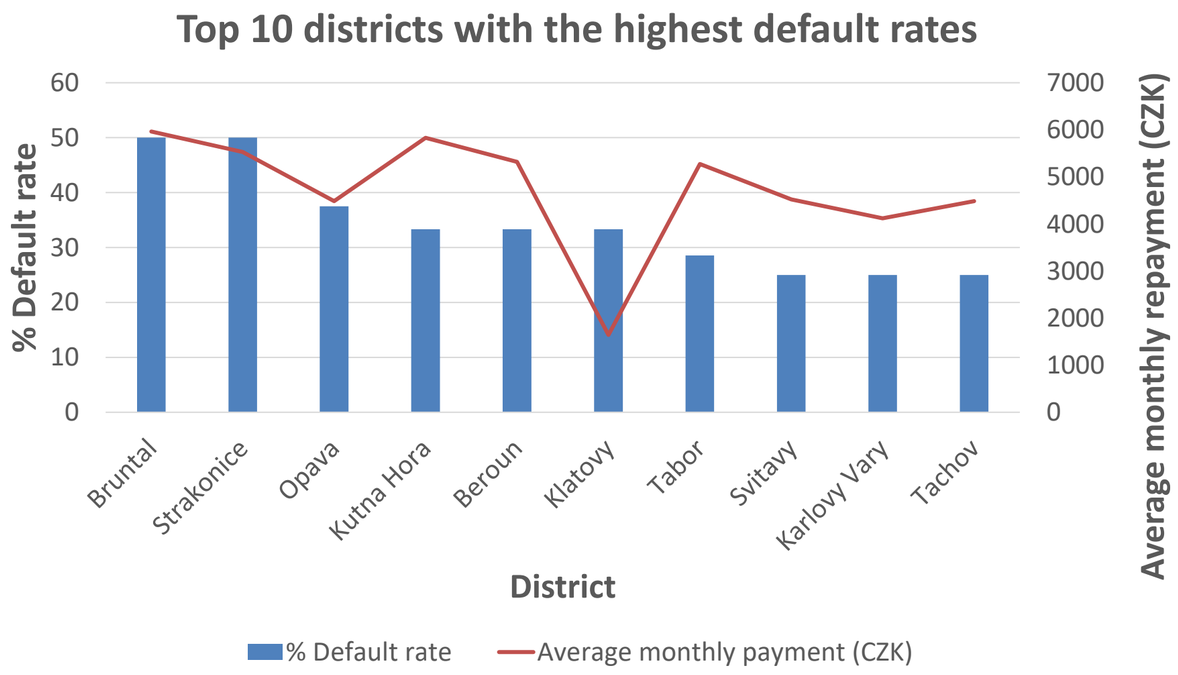


Reference tables:
Loan – 682 rows
Client – 5369 rows

There is a proportional relationship between loan amounts and default rates

- As the loan amount increases the default rate increases because this will result in higher monthly repayments which have been shown to increase the risk defaulting
- The districts with the highest default rates also have the highest monthly repayment rates, further validating the finding that higher monthly repayments increase the risk of defaulting.

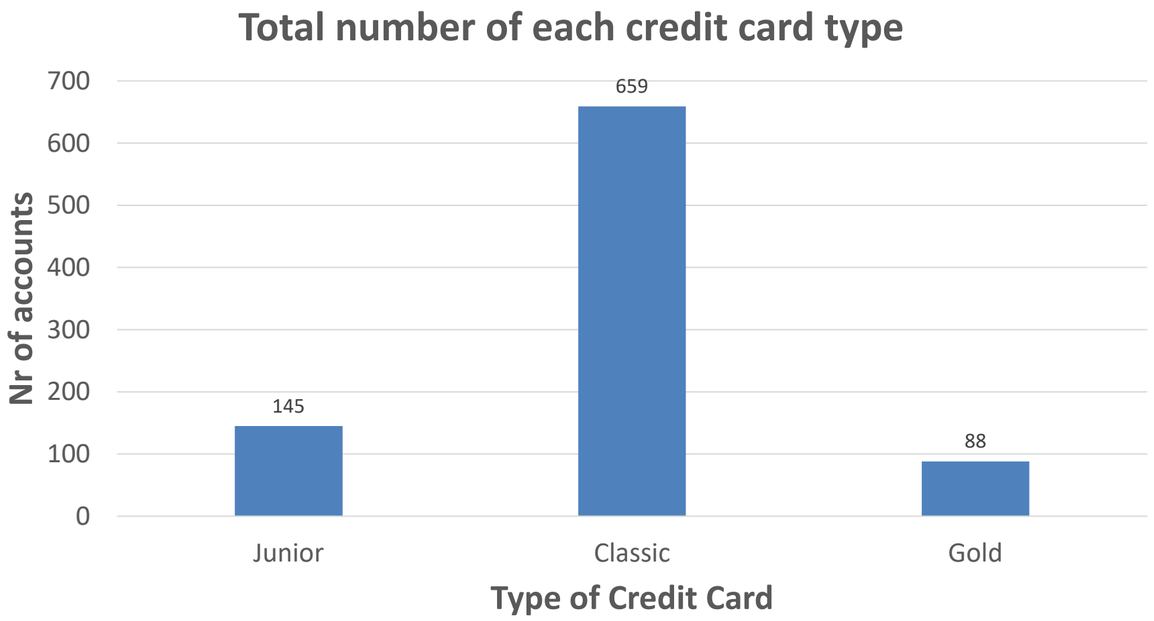


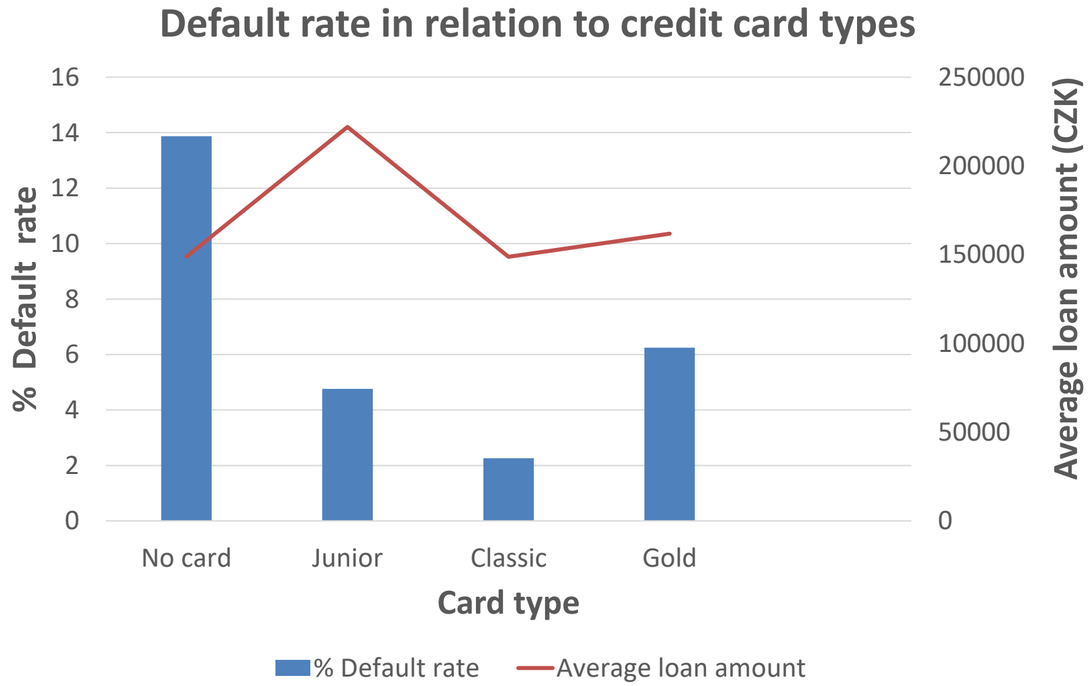


Reference tables:
Loan – 682 rows
Demograph – 77 rows

The Classic credit card is the most used and has the lowest default rate compared to other credit card types and non-credit card holders

- Classic credit card users are financially and credit-savvy consumers → Have the lowest average loan amount and default rates
- Junior credit card users have the highest average loan amounts and 2nd lowest default rates
- Consumers without credit cards, have the 2nd lowest average loan amount, yet still have the highest default rates

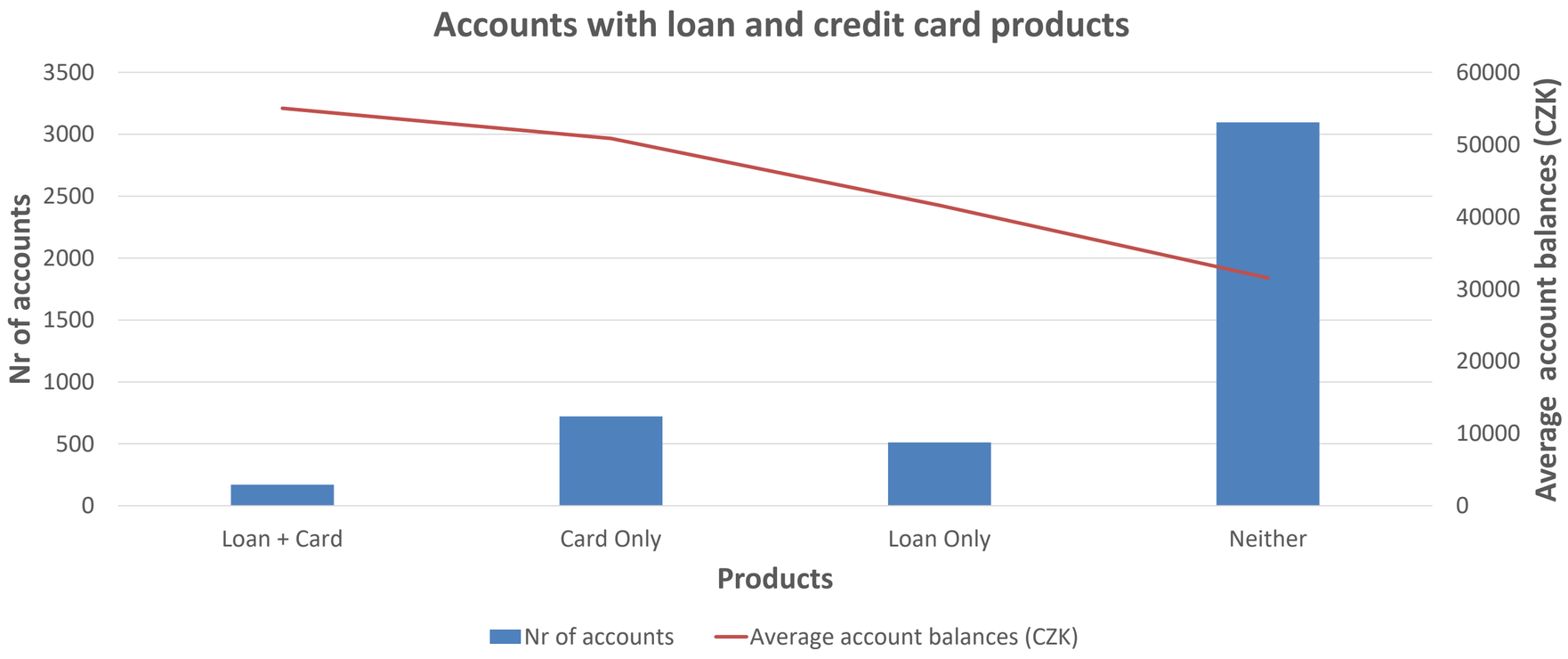




Reference tables:
Client – 5369 rows
Credit cards – 892 rows

Furthermore, clients with loans and credit cards are the least but on average have higher account balances compared to those without these products

- It was interesting to note that majority of clients have neither loans nor credit cards but had the lowest average account balances.
- Clients with both loans and credit cards represent the smallest group, yet they hold the highest average account balances → These are high value consumers who know how to use credit



Reference tables:
Client – 5369 rows
Credit cards – 892 rows
Loan – 682 rows

We have learnt the following lessons about loan defaults

1. Risk to default increasing when the loan amount increases
   ↑ loan amount = ↑Risk to default

2. High monthly repayment in relation to the client's salary

   Loan durations that are longer than 24 months

3. Regions can impact default rate.
   The following should be taken into consideration in regions:
   - Average salary earned
   - Unemployment rate

4. Clients with credit cards have the lowest default rates →
   Credit card usage is a powerful proxy for financial discipline and creditworthiness

The dataset is from 1999 but it shows the importance of a credit score…

Clients who actively manage a credit card are demonstrating the exact behaviour the bank wants in a loan customer

The bank should actively use credit card ownership as a positive signal in its loan underwriting model, and consider offering credit cards to loan applicants as a way to build their credit profile before providing larger loans

January 2026

Thank You

Czech Bank Analysis

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